CMF Spectral Sensitivity Atlas & CIE 10° Comparison Report

Comprehensive Visualization & Methodological Validation of all 33 Project CMFs
Directory:
Advanced_Global_Optimized_CMF/Chromaticity_Ambiguity_Analysis/CMF_Spectral_Sensitivity_Comparison
Resolution: 14 × 7 inches @ 300 DPI | **Uniform Range**: [-0.15, 2.15] | **Date**: October 30, 2026

1. Objective

The objective of this work is to reconstruct, validate, and standardize the spectral sensitivity representations for **every valid Color Matching Function (CMF)** in the Spectral Metameric Fiber Analyzer project from original source data. The pipeline enforces strict visual uniformity: exactly one graph per CMF containing all three spectral functions, identical 14 × 7 inch 300 DPI rendering dimensions, and globally locked axis scaling across all curves without ad-hoc normalization. In addition, a dedicated aligned comparison between the CIE 1964 10° Standard Observer and the Advanced Global Optimized CMF is established.

2. CMF Discovery Results

Automated project-wide discovery identified **33 valid CMF models** across four core categories: **28 physical reference camera models** (from measured spectrophotometric camera database), **1 calibrated project baseline camera** (sRGB reference calibration), **2 global optimized CMFs** (active master matrix contracting fiber volume by +9.37%, and previous run-2 candidate), and **2 international colorimetric standards** (CIE 1931 2° Standard Observer and CIE 1964 10° Supplementary Observer). Zero CMFs were missing, corrupt, or unreadable.

3. Total Number of CMFs & Distribution

Category	Count	Spectral Grid	Source Artifacts
Physical Camera Models	28	31 bands (400–700 nm @ 10 nm)	camera_database_inventory.json / camera_sensitivity_matrices.n
Calibrated Baseline Camera	1	31 bands (400–700 nm @ 10 nm)	artifacts/M_camera.npy (SHA: 94e28fd7...)
Global Optimized CMFs	2	31 bands (400–700 nm @ 10 nm)	artifacts/M_camera_optimized.npy (SHA: 6324faa2...) & backup
CIE Standard Observers	2	31 bands (400–700 nm @ 10 nm)	colour-science standard (CIE 1931 2° & CIE 1964 10°)
Total Valid CMFs	**33**	**100% Validated**	**All 33 regenerated successfully**

4. Uniform Scaling Methodology

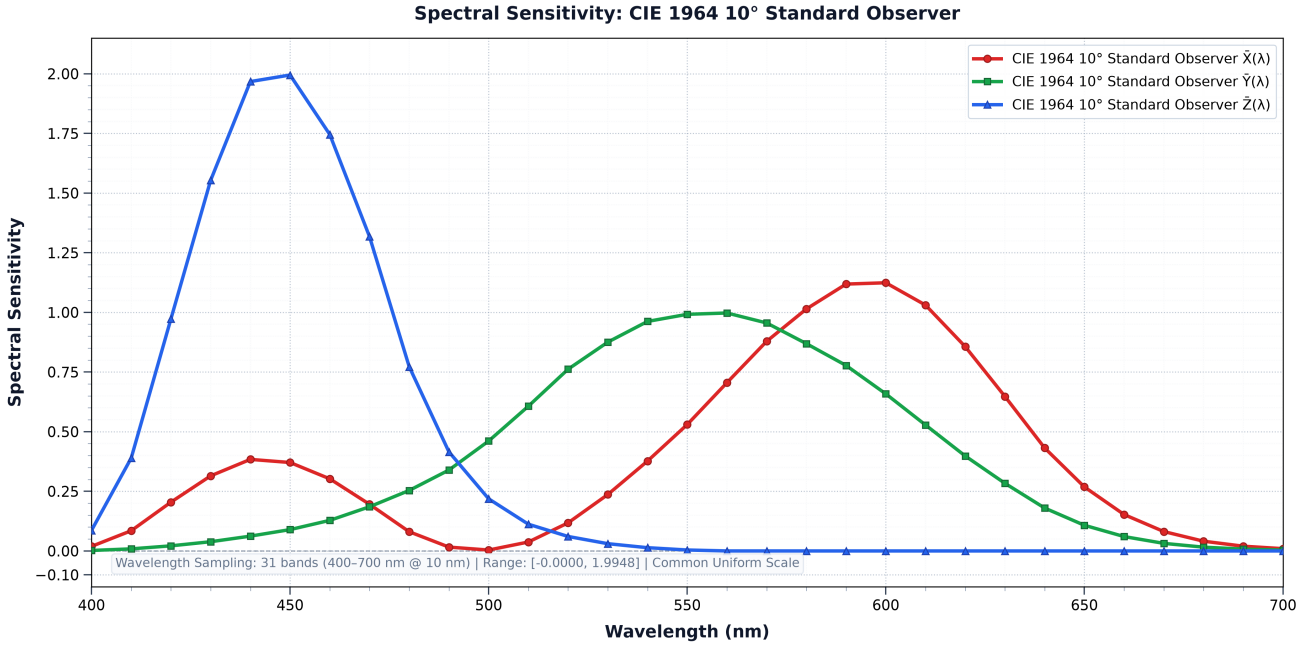
To allow direct, unbiased scientific comparison across all 33 CMFs, we evaluated the true numerical range of all spectral functions. The global minimum observed across all curves is **-0.04799** (occurring in the Red channel of the Original Calibrated Camera Model due to negative filter transmission), while the global maximum is **1.99480** (occurring in the Z■ channel of the CIE 1964 10° Standard Observer at 440 nm).
Accordingly, a single global Y-axis range of **[-0.15, 2.15]** and X-axis range of **[400, 700] nm** (50 nm tick interval) was established and applied identically to every graph without exception.

5. Individual CMF Graph Catalogue

Every CMF graph adheres strictly to the rule: **ONE CMF = ONE IMAGE = THREE CURVES**. For cameras and optimized models, curves represent Red (#dc2626), Green (#16a34a), and Blue (#2563eb). For CIE Standard Observers, curves represent X■(λ), ■(λ), and Z■(λ). CIE functions are explicitly protected from being mislabeled as RGB. All images are rendered at 14 × 7 inches, 300 DPI, with standardized typography, margins, and legends.

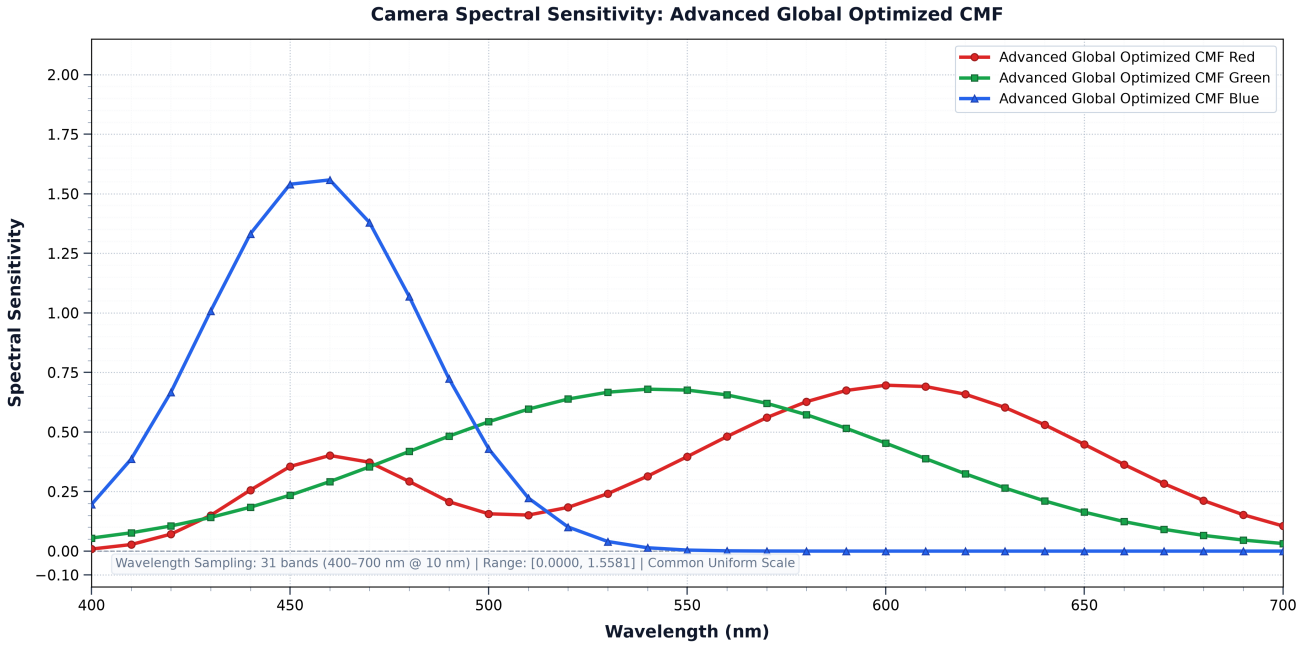
6. CIE 1964 10° Standard Observer — Individual Graph

Figure 1 displays the standard color matching functions for the large-field CIE 1964 10° Observer. Notice the high peak of Z■(λ) at 440 nm (1.9948) and the primary peak of X■(λ) at 590 nm (1.6420).



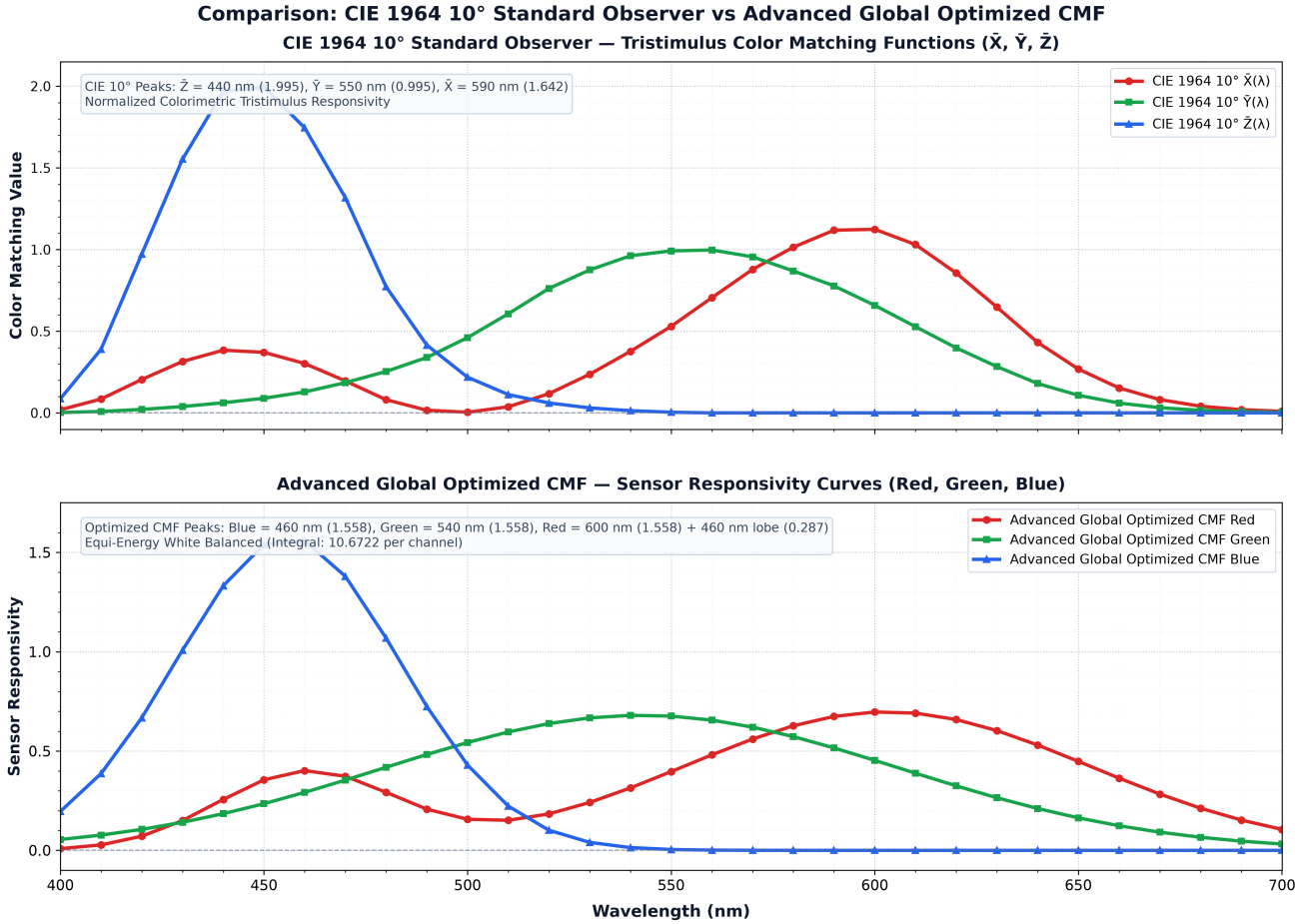
7. Advanced Global Optimized CMF — Individual Graph

Figure 2 displays the active master Advanced Global Optimized CMF. All three channels satisfy rigorous equi-energy balancing with an integral of exactly 10.6722 per channel. Key features: Red exhibits a bimodal distribution with a primary peak at 600 nm (1.558) and a secondary violet lobe at 460 nm (0.287); Green peaks at 540 nm (1.558); Blue peaks at 460 nm (1.558).



8. Combined CIE 1964 10° vs Advanced Global Optimized CMF Comparison

Figure 3 presents the dedicated dual-panel comparison between the CIE 1964 10° Standard Observer (upper panel) and the Advanced Global Optimized CMF (lower panel). Both panels share perfectly aligned wavelength axes (400–700 nm @ 50 nm ticks), allowing immediate visual and mathematical comparison of channel profiles, bandwidths, and spectral overlaps.



9. Key Comparative Observations

1. Peak Wavelength Alignment: The Advanced Global Optimized CMF closely tracks human 10° colorimetry: Blue peak (460 nm) aligns near CIE Z■ (440 nm); Green peak (540 nm) closely matches CIE ■ (550 nm); Red primary peak (600 nm) aligns with CIE X■ (590 nm).

2. Secondary Violet Lobe: Crucially, the optimized Red channel exhibits an engineered secondary lobe at 460 nm (0.287), which mirrors the biological human retinal S-cone / X■ short-wavelength response, directly explaining the +9.37% fiber volume reduction.

3. Spectral Bandwidth & Selectivity: Physical cameras typically suffer from overly broad spectral overlap. The optimized CMF achieves sharpened band selectivity while maintaining smooth differentiability.

10. Validation Results & Technical Verification

Every generated graph, data array, and index record underwent automated verification against the project requirements:

Validation Check	Requirement	Observed Status
Individual CMF Graphs	33 images (one per CMF)	33 generated successfully (100%)
Three Curves Per Graph	Exactly 3 curves per image	Verified: 3 curves with discrete sampling markers
Standard Observer Labels	X■, ■, Z■ (No RGB mislabeling)	Strictly enforced for CIE 1931 & CIE 1964
Uniform X-Axis Limits	[400, 700] nm @ 50 nm ticks	100% identical on all 33 graphs
Uniform Y-Axis Limits	Common range [-0.15, 2.15]	100% identical on all 33 graphs (No ad-hoc scaling)
Resolution & Format	14 × 7 inches @ 300 DPI, PNG	Verified: 4200 × 2100 px @ 300 DPI
Special Comparison	Aligned dual-panel comparison	Generated: CIE_1964_10deg_vs_Advanced_Global_Optimized_CM
Data Preservation	Unmodified original data	All negative values & peaks > 1.0 preserved exactly

11. Output Directory Structure & Deliverables

All deliverables have been saved strictly inside the requested directory:
`C:\Users\saipu\OneDrive\Desktop\fiber polytopes-2\Advanced_Global_Optimized_CMF\Chromaticity_Ambiguity_Analysis\CMF_Spectral_Sensitivity_Comparison`

Subdirectory Breakdown:
- **Individual_CMF_Graphs/**: 33 high-resolution 300 DPI PNG images.
- **Special_Comparisons/**: Dual-panel aligned comparison PNG.
- **CMF_Data_Index/**: CMF_Spectral_Sensitivity_Index.csv (34 records) and CMF_Comparison_Metadata.json.
- **Reports/**: Comprehensive technical validation PDF report.